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Bio-inspired navigation systems for robots

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Bio-inspired path integration and neural mapping enable robots to navigate autonomously in global positioning system (GPS)-denied environments such as disaster zones, space and deep oceans.

In a catastrophe zone, a search-and-rescue robot must be able to navigate unpredictable and complex terrains without the help of a global positioning system (GPS). Similarly, a Mars rover charting alien terrain should be able to navigate without human input. However, conventional robots have trouble in uncertain, GPS-denied situations; they can become disoriented easily, run out of battery or stumble over unforeseen obstructions[1]. Additionally, conventional navigation systems are energy intensive, computationally demanding and susceptible to sensory degradation. These challenges can be addressed by looking to nature: animals have diverse innate navigation mechanisms that can inform robot design.

Bio-inspired designs that mimic nature's navigation strategies can enable operation without GPS. For example, conventional simultaneous localization and mapping (SLAM) systems rely on computationally expensive algorithms and precise, energy-intensive sensors such as light detection and ranging (LiDAR), yet are susceptible to drift and sensory degradation in chaotic environments. Bio-inspired design improves on SLAM by mimicking the efficient neural processes of animals; for example, using event-based vision sensors that mimic an animal's responsive eyes, neuromorphic chips for low-power 'neural' computation and algorithms such as hierarchical temporal memory that dynamically fuse multi-sensory data. By replicating the brain's ability to process only essential information and weight sensor reliability in real time, these systems can minimize computational overhead. This event-driven, efficient processing is the key to their energy savings, enabling robots to navigate complex, GPS-denied terrains for extended durations on limited power.

Navigation strategies

Bio-inspired systems in combination with state-of-the-art sensors can be deployed in real-world applications such as disaster response, planetary exploration and deep seas[2–4]. For example, desert ants use 'path integration', a navigation strategy involving step counting while tracking the sun's position, analogous to a hiker using a pedometer and compass. This approach has been implemented in field robots such as the AntBot[5], which uses optical encoders for step counting and a celestial compass for solar alignment. In featureless terrain, such systems reduce positioning errors by 40% compared to inertial measurement unit (IMU)-based odometry. AntBot has been used for autonomous data collection in GPS-denied agricultural fields, navigating between crop rows using its celestial compass. Additionally, rodents build mental 3D maps using 'place cells' in their brains. This ability is mimicked in robots through 'neural odometry', a brain-inspired algorithm that instantly updates maps when obstacles appear. For example, the RatSLAM robot[6] performed autonomous exploration and mapping of collapsed structures, reducing drift by 58% compared to conventional LiDAR-based SLAM systems in rubble-filled disaster zones[6]. Furthermore, birds navigate long distances using Earth's magnetic field and now drones equipped with quantum sensors can detect these invisible forces, enabling them to navigate underground or underwater without GPS[7].

Animals also excel at integrating multiple senses to survive, and this multi-sensory fusion can be mimicked in robots. Robots blend various sensors: LiDAR for distance measurement, cameras for visual data and magnetic sensors for orientation (Fig. 1). These inputs are fused through bio-inspired neural networks — such as hierarchical temporal memory models — that mimic the brain's ability to weight sensor reliability in real time, dynamically switching between modalities when one fails. For example, if dust blinds a camera, the system shifts to other sensors. Field tests with ground robots showed this multi-sensory approach enabled them, on average, to recover their navigation function 60% faster than those relying on a single sensor[8]. Additionally, swarm coordination, such as ants' chemical trails, inspired digital 'pheromones' — virtual signals that robots broadcast to share paths wirelessly[9]. This enables 25% faster search coverage than by individual robots. In field tests, 50 wheeled robots located thermal targets in simulated disasters within 4 minutes using ant-colony coordination[9]. The 'hybrid bio-inspired framework' (a modular system combining animal strategies contextually) uses ant-like sun navigation in open fields and rodent-style mental mapping in cities. For group coordination, the framework can incorporate ant-inspired swarm rules. For example, agricultural drones using this framework achieved a 90% success rate in autonomous navigation through orchards without collisions. Similarly, as part of the hybrid bio-inspired framework, Mars rovers can combine bird-inspired celestial navigation for maintaining positioning during sandstorms via stellar tracking with rodent-inspired cognitive mapping for exploring complex geological features such as caves and lava tubes[10].

Outlook

The future of robotics promises smart and capable machines by blending animal behaviours with artificial intelligence. Neuromorphic chips engineered to mimic the event-driven processing of ant brains, wherein only relevant circuits activate on demand, could reduce energy consumption by up to 90%, enabling micro-bots to operate for weeks on a single charge[2]. Additionally, drones using reinforcement learning algorithms can master long-haul flights by analysing bird migration patterns. In space exploration, hybrid systems combine ant-inspired pathfinding for dune navigation with rodent-like cognitive mapping for cave exploration, enabling autonomous Mars rovers[10]. However, hurdles in scalability, sensory overload and ethical considerations limit the widespread adoption of bio-inspired navigation systems.

In terms of scalability, coordinating more than 100 robots causes communication gridlock (n^2 interactions), with field tests failing

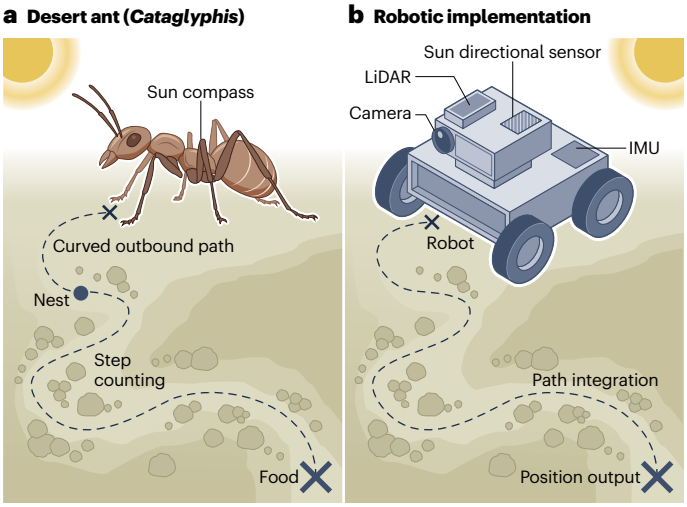


Fig. 1 | Bio-inspired path integration. The figure demonstrates the direct translation of a desert ant's biological navigation mechanism to a robotic system. **a**, The desert ant (*Cataglyphis*) performs path integration by using an internal 'sun compass' for direction and 'step counting' for distance, enabling it to calculate a direct path back to its nest after a winding search for food with accuracy (<1° directional error over 1 km). **b**, A robotic implementation mimics this strategy. A 'sun directional sensor' replaces the sun compass, and an inertial measurement unit (IMU) and wheel encoders perform 'path integration' to track movement, analogous to step counting. This enables the robot to estimate its position ('position output') in global positioning system (GPS)-denied environments. Additional sensors such as a camera and light detection and ranging (LiDAR) provide data for obstacle avoidance.

beyond 50 units. Therefore, decentralized, ant-inspired local interaction rules based on stigmergy could be implemented to enable the coordination of thousand-unit swarms[6]. Sensory overload also disrupts performance; during the DARPA Subterranean Challenge, a competition focused on robotic exploration of underground environments, robots froze when flooded by multi-sensor data from suites often comprising LiDAR, stereo cameras, thermal imagers and IMUs[2,7]. However, dragonfly attention mechanisms, such as a 'centrifugal visual filter' that suppresses irrelevant background clutter and 'selective attention neurons' that lock onto a single target amid distractions, can enable robots to prioritize critical inputs and avoid overload[7]. Ethics also comes into play, particularly with the potential weaponization of robot swarms. This calls for the development of 'ethical kill switches' and global regulations to prevent misuse of swarm technologies[7]. Ethical governance gaps could adopt honeybee consensus algorithms for swarm accountability[9]. Resolving these limitations could potentially enable 5-minute survivor searches by ant-coordinated robot swarms in disasters, autonomous coral reef restoration using fish-inspired drone builders and radiation-shielded lunar bases constructed by rodent-like mapping rovers[8,10]. Critical enablers for these future applications include neuromorphic chips, which mimic the brain's event-based processing to reduce energy consumption by up to 90% compared to traditional processors, and the development of global artificial intelligence ethics frameworks to ensure responsible deployment[2,8,10].

Conclusion

The advancement of bio-inspired robotics brings pressing ethical challenges that demand proactive solutions. For example, agricultural drones, although beneficial for crop monitoring, could encroach on privacy if they collect off-farm property data without consent, a concern that is already debated in the field of precision agriculture. Autonomous rescue robots face accountability gaps when making unsupervised life-or-death decisions, leaving responsibility ambiguously divided between developers and operators. Similarly, the dual-use potential of swarm technologies, which are designed for search and rescue but are adaptable for combat applications, further complicates their ethical deployment. To mitigate risks, the field is implementing safeguards: mandatory emergency kill switches for human override of malfunctioning swarms, general data protection regulation-compliant data systems that anonymize sensitive information in real time and even environmental protections such as high-frequency sound emitters to minimize ecological disruption from underwater robots. These measures align with the guiding principle that responsible innovation requires ethical forethought. Global collaboration among engineers, ecologists and policymakers is crucial to establish frameworks ensuring these technologies prioritize safety, privacy and sustainability. By addressing these challenges now, we can harness bio-inspired robotics' potential while safeguarding societal values.

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Competing interests

The authors declare no competing interests.